

GPStroke: A Walking Promotion Method using GPS Art with Multiple Walking Paths

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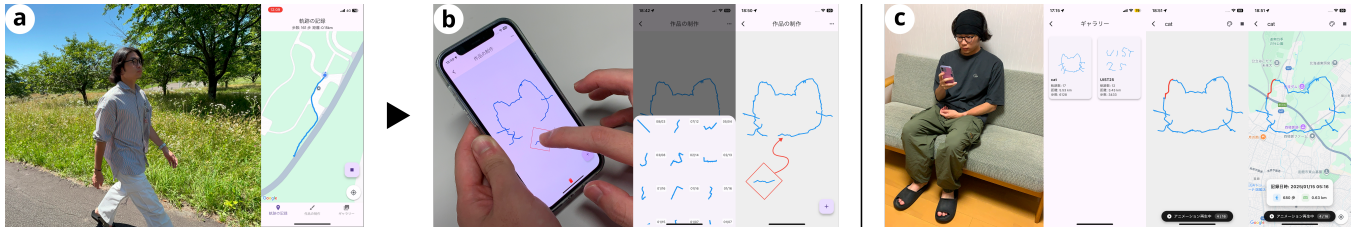


Figure 1: The GPStroke outline. (a) A user can record walking paths as "strokes" using smartphone GPS during daily movements. (b) The user can later create a GPS art by combining the walking paths on the smartphone. (c) The user can browse the art pieces and additional information (e.g., surrounding map of each path).

Abstract

Walking is known to be effective in maintaining health. Conventional walking promotion methods require dedicated exercise time, which is particularly difficult for busy people. Therefore, we propose a system called "GPStroke" that collects fragmented daily walking paths as "strokes" and later combines them to create GPS artworks to easily incorporate walking activities into daily life. GPS art is a creation method that utilizes GPS logging functions of smartphones to visualize actual walking paths on a map as line drawings. However, conventional GPS art requires planning a route in advance and continuous walking for a long period. With GPStroke, users can record paths during daily movements, such as commuting and shopping, and later use them as strokes to create artworks. By separating walking activities from artwork creation, GPStroke realizes a mechanism that allows people to continue walking activity by utilizing everyday travel. This paper describes the system design and implementation of GPStroke.

CCS Concepts

• Human-centered computing → Interaction design.

Keywords

GPS art, behavior change, walking

ACM Reference Format:

Etsuki Omoto and Koji Tsukada. 2025. GPStroke: A Walking Promotion Method using GPS Art with Multiple Walking Paths. In *SIGGRAPH Asia*

2025 Posters (SA Posters '25), December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771490>

1 Introduction

Physical activities, including walking have been reported to reduce the risk of non-communicable diseases and improve mental health[World Health Organization 2024b]. WHO recommends that adults engage in at least 150 min of moderate-intensity exercise per week; however, a recent survey[World Health Organization 2024a][Strain et al. 2024] showed that approximately 31% of the world's adults do not meet this recommended level, which has increased from 23.4% in 2000. Research on barriers to physical activity has consistently reported "lack of time" as the most important factor[Cavallini et al. 2020][Koh et al. 2022][Reichert et al. 2007]. This time constraint stems from the difficulty in balancing daily work and exercise. Given this background, walking promotion methods that can be practiced with limited use have been proposed. Conventional walking promotion applications, such as Fitbit[Fitbit Inc 2024] and Apple Health[Apple Inc 2024], record step counts and moving distances. However, these mainly focus on numerical feedback and lack elements to make walking activities enjoyable. Gamification-based methods such as Pokémon GO[Niantic, Inc 2016], have been proposed. While these show certain effects in motivating walking, they remain difficult for people with time constraints to practice because of the requirement of a certain specific time for game playing.

This study focused on methods to promote joyful walking activities using GPS art. GPS art is an activity that visualizes walking routes as line drawings on maps. This method has the potential to transform movement into a creative activity, thereby increasing motivation for walking. Recently, systems that support the creation of GPS art have been proposed. Strava Art[Strava 2024] is a platform on which users can enjoy their exercise records as artworks by drawing running and walking routes on maps. Waschke et

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SA Posters '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771490>

al.[Waschk and Krüger 2018] proposed a system that automatically generates walking routes on maps using sketches and shapes as input, aiming to reduce the labor involved in route planning, which was traditionally performed manually. However, conventional GPS arts using these support systems typically require the decision of a theme for the artwork in advance, strictly planning movement routes, and drawing the work in one continuous stroke. Although this is a creative and attractive activity, it requires dedicated time blocks for planning and walking, making it difficult to incorporate casually into daily life.

To solve the above challenges, this paper proposes "GPStroke" a system that collects and accumulates fragmented movement paths from daily life as "strokes" and later combines them to create GPS art.

2 GPStroke System

In this section, we describe the GPStroke system, which addresses the aforementioned challenges. GPStroke enables users to create GPS art by combining multiple walking paths recorded during their daily activities. The features of this approach are as follows:

- Utilizing walking paths from daily activities (e.g., commuting and shopping) as creative materials.
- Creating artworks later with the walking paths, which does not require a certain time for exercise.

Through this approach, we aim to motivate and support busy people to continue walking through creative experiences.

GPStroke is implemented as a mobile application compatible with iOS/Android using the Flutter framework and consists of the following three functions:

- (1) Path recording: record and save movement paths using smartphone GPS
- (2) Artwork creation: create artworks by combining multiple recorded paths
- (3) Artwork browsing: browse created artworks

For technical implementation, GPS tracking dynamically adjusts location update intervals based on user movement speed, achieving accurate path recording while minimizing battery consumption. The artwork creation interface enables users to place, move, and rotate strokes using touch gestures, allowing the free combination of multiple walking paths. For data management, the system employs both local storage and a cloud server (Firebase Firestore), managing path data and artwork information online to enable data sharing across devices.

2.1 Path Recording

Users can record walking paths as "strokes" using smartphone GPS simply by tapping the record button during daily movements such as commuting or shopping (Figure 1a). Recorded strokes are saved along with information such as date and moving distance, and accumulated for later artwork creation.

2.2 Artwork Creation

This is the core function of the system. Users select materials to use from a list of accumulated strokes and place them on the canvas (Figure 1b). Each stroke placed on the screen can be manipulated

through drag operations and rotated using pinch gestures. Users can draw inspiration from the shapes of recorded paths, for example, viewing a "curved path" as an "animal's ear," and combine strokes like a puzzle to create a single artwork.

2.3 Artwork Browsing

Completed artworks can be saved with titles. By browsing their own works, users can visualize their accumulated walks as artworks. Furthermore, the system includes an animation function that replays how completed works were created (Figure 2). After displaying the entire completed work, the walking paths that compose the work are highlighted one by one in the order in which they were recorded. Along with highlighting each trajectory, information such as maps, recording date, number of steps, and distance is also presented, allowing users to understand the process of artwork creation (Figure 1c).

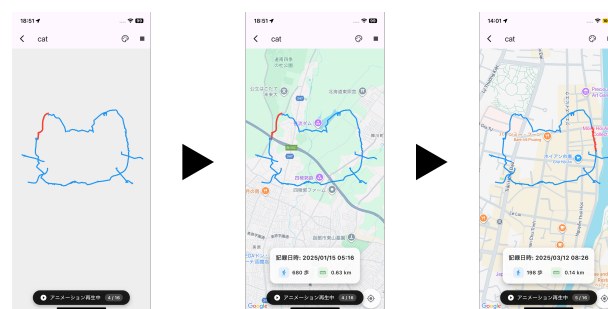


Figure 2: Animation function

3 Conclusion and Future Work

This paper proposed "GPStroke," a system that creates GPS art by combining fragmented daily walking paths. This system provides a new approach that allows people with time constraints to continue walking activities enjoyably by utilizing daily movements as creative materials. In the future, we plan to implement social functions that enable created works to be shared with friends, as well as route suggestion functions that encourage active walking based on "desired trajectory shapes." We will also verify the effectiveness of the system through long-term evaluation.

Acknowledgments

This work was supported in part by the FY2025 Hokkaido IT Creator Discovery and Development Program, administered by Shinsetsu, a general incorporated association.

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